

Design the Internet, part 2

Assignment #2

Software Systems Architecture

1MEIC02 T3

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1. Describe two crucial decisions (excluding reliability, routing and resolution)

There are many other crucial decisions talked about in the article, such as: assuming failure is the normal case, rough consensus and running code, the singularity of the internet protocol and infrastructural autonomy.

We will describe and explain the following two: assuming failure is the normal case and rough consensus and running code.

1. Assuming failure is the normal case: This crucial decision is mentioned in Page 72 (9 of the article) and its core meaning is that instead of designing a system that expects every operation to be perfect and enters a special “recovery mode” when something fails, the Internet’s design treats component failure as a routine event.

By doing this, it allows the system to scale massively, because handling failures efficiently is a key scalability requirement. For example, routing protocols recalculate paths the same way whether it's for a periodic update or link failure, and transport protocols expect and handle packet loss frequently rather than treating it as an error. This philosophy was later used also in modern hyperscaled infrastructures, such as Google’s.

2. Rough consensus and running code: This secret to Internet’s success is mentioned in Page 72 (9 of the article) and it refers to the community-driven design process used by the Internet’s creators. They denied formal and rigid design committees, which were common in the telephony industry, in favor of small groups building working prototypes and seeking community agreement. Using this approach, it shared a vision of a unified communication platform and allowed for pragmatic, functional growth rather than theoretical perfection.

2. “Why instead of how” - two examples where “why” gave extra insight into the Internet

Throughout the article, the authors prioritize the design rationale, the 'why', over the technical implementation, or the 'how.' This approach offers a valuable perspective; while we are often taught how a system functions or how to build it, we rarely consider the underlying reasoning behind those choices. By focusing on the 'why,' the authors encourage a more critical way of thinking, framing the Internet’s creation as a series of specific problems that could have been solved in various ways, but were ultimately shaped by intentional architectural decisions.

Two particular examples stood out to us, offering new insights into how the Internet functions and forcing us to reconsider our own design choices. These examples demonstrated that understanding the reasoning behind a protocol is just as important as knowing how it is implemented:

1. Modesty: The Internet was intentionally designed to be 'modest' rather than following the ambitious, high-performance model of the traditional telephony system. While at first glance, a simple service model, which makes no guarantees about data delivery, might seem like a limitation, but the rationale behind it was strategic. By adopting a low-end service model, the creators ensured that the Internet could interconnect almost any type of

hardware. This modesty in performance requirements was the key to its success: because it required so little from the underlying infrastructure, it could achieve a scope and speed of global deployment that a 'perfect' or more complex system never could have reached. This gave us the insight that prioritizing universal compatibility over guaranteed performance is what allowed the Internet to become a ubiquitous, common platform.

2. Narrow Waist: A second insight that changed our perspective was the deliberate decision to have only one protocol at the Internet networking layer, the “narrow waist”. While the Internet encourages a diverse range of choices at the application and physical layers, it strictly enforces a universal protocol (the Internet Protocol, IP) at Layer 3, as shown on *Figure 1*. The rationale, the “why”, is that this single, common interface acts as a universal translator. It creates a decoupling effect: application developers can create new apps without knowing if the data travels over fiber or satellite, and hardware engineers can build new physical layers without knowing what type of data they will need to carry. This gave us the insight that by strictly limiting choice at one strategic point (the waist), it actually creates the freedom for infinite innovation everywhere else.

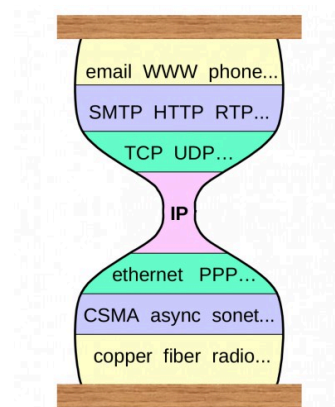


Figure 1. The Internet's "Narrow Waist" Architecture

3. The meaning behind the title: *“Extracting The Essential Simplicity of the Internet”*

The authors want to show through the title of the article that the Internet is built on a few very basic and smart ideas. Most networking course students get lost in an “alphabet soup” of confusing technical names and messy rules, but the authors argue that the real beauty of the Internet is buried under those details. By “extracting” this simplicity, they are pulling out the core plan that has remained almost the same since 1983, even while the technology around it changed completely.

This simple approach comes from a famous historical design choice called the **End-to-End Principle**. In the past, telephone companies built “smart” networks that tried to manage everything for the user, which made them very rigid and hard to change. The Internet’s creators did differently: they focused on keeping the center of the network as simple as possible and moving the “smart” work to the computers at the ends of the connection. This historical choice was the key to make the Internet so successful, as it allowed new inventions like video streaming and social media to be created by users without needing permission from the network itself

4. Comparison with our design and lessons

Similarities

Our design and the real Internet share a similar big picture strategy for connecting the world, such as, both use a hierarchy where high-speed “backbones” (our “Super-Fiber” and “Continental Anchors”) move data long distances, while local “Navigators” handle the final delivery. Both systems also use a best-path discovery process. We use “Hop Counts” to find the nearest exit to the global mesh, which is very similar to the “distance-vector” logic used in real world routing to find the shortest path between routers.

Additionally, both systems break data into small units called packets to achieve efficient resource utilization during communication, and use hierarchical addresses. The [Anchor].[Regional].[Navigator].[User] format is conceptually similar to how IP addresses identify specific networks and host interfaces within those networks.

Differences

The primary difference lies in the level of complexity built into the network infrastructure. While both designs rely on end users to handle data errors, our design is significantly more ambitious than the actual Internet’s modest “best-effort” model. The proposed design includes a “Priority System” to actively manage emergency data, whereas the actual Internet does not offer any kind of performance guarantees for such cases.

Furthermore, our approach adds an extra layer of network functionality through “Backup Parents,” providing alternative routes if a primary connection fails, allowing users to instantly “hop” to a backup Navigator to prevent the connection from breaking in the first place, while the real Internet assumes that dropped packets are a normal part of life and leaves recovery to the host devices.

The Lesson

The most important lesson here we learned is, a **“dumb” network is often more “intelligent” for growing at a global scale**. By keeping the center of the network as simple as possible, the Internet's creators made a system that was flexible enough to run on any hardware and support any new app without needing a redesign.

Our design tries to do a lot for the user, but the real Internet succeeded by doing very little at the core. This “essential simplicity” is what allowed the Internet to remain almost unchanged since 1983 while still supporting everything from simple emails to high-definition video.